

ICCT COLLEGES FOUNDATION INC

College of Engineering and Digital Technology

PROJECT NEXUS

Parking Slot Reservation & Management System

A Capstone Project Presented to the Faculty of the
College of Engineering and Digital Technology,
In Partial Fulfillment of the Requirements for the Course
Software Design

Presented by:

Alba, Paul Vincent - Project Manager

Amaba, Arjohn - Database Administrator (DBA)

Vergara, John Michael - Lead Backend Developer

Balean, Kristiancarl - UI/UX Lead Designer

Agcambot, Raphael Francisco - Technical Writer & UML Architech

Velasco, Christianmark - UAT Coordinator & Pitch Presenter

Basence, Francis - Quality Assurance (QA) Engineer

Alvarez, Faith - System Analyst

Moran, Adriel John - Lead Frontend Developer

Belardo, John Carlo - DevOps & Development Specialist

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Approval Sheet

This project, titled "[System Title]", prepared and submitted by [Team Members], in partial fulfillment of the requirements for the course Software Design, has been examined and is recommended for acceptance and approval.

Adviser

Accepted and approved in partial fulfillment of the requirements for the degree of Master of Arts in Education major in Physical Education and Sports.

JONATHAN C. NAVARRO, MIT, PhD
PANEL

DENVER PANTE MIT, PhD
PANEL

Acknowledgment

[Optional. A brief note from the team thanking advisers, panel members, and contributors. One page maximum.]

Abstract

A concise (250–300 word) summary covering: the problem addressed, the proposed solution, the methodology/tech stack used, key results from testing, and the deployment outcome. The abstract is written LAST, after Chapter 4 results are final, even though it appears first in the paper.

Suggested keywords line beneath the abstract, e.g.: Keywords: campus system, Agile, MVC, [your tech stack].

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CHAPTER 1 - THE PROBLEM AND ITS BACKGROUD

1.1 Introduction

Finding an available parking space can be difficult, especially in places where many vehicles arrive at the same time. Drivers may spend several minutes going around the parking area just to find an empty slot. This can result in wasted time, traffic within the parking area, and confusion regarding which spaces are already occupied. In some parking areas, the availability of spaces is still monitored manually, making it difficult to provide accurate and updated information.

The proposed **Parking Slot and Management System** is designed to help improve the way parking spaces are monitored and managed. The system will allow users to view the availability of parking slots and determine which spaces are occupied or vacant. It may also provide administrators with a way to monitor parking activity and manage information about parking slots.

Through the use of a computerized management system, the proposed application aims to make parking more organized and convenient. Instead of relying entirely on manual monitoring, the system can provide updated parking information that can help drivers find available spaces more efficiently.

The proposed system will focus on developing a simple and user-friendly solution that can be applied to a selected parking area. It is intended to demonstrate how technology can be used to improve parking management and reduce common problems associated with locating and monitoring available parking spaces.

1.2 Statement of the Problem

Parking management can become challenging when the number of vehicles increases and available parking spaces are limited. Without an organized system, drivers may have difficulty identifying vacant slots, while parking administrators may find it difficult to keep track of occupied and available spaces.

The proposed study aims to address the following problems:

1. How can the system provide users with information about available and occupied parking slots?
2. How can the system help reduce the time spent by drivers searching for an available parking space?
3. How can parking administrators efficiently monitor and manage the status of parking slots?
4. How can the system maintain organized records of parking slot usage?
5. How can a computerized parking management system improve the overall organization and convenience of a parking area?

1.3 Objective of the Study

1.3.1 General Objective

The general objective of this study is to develop a Parking Slot and Management System that can monitor parking slot availability and assist in the efficient management of parking space***s.***

1.3.2 Specific Objectives

Specifically, the study aims to:

1. Develop a system that displays the current status of parking slots as available or occupied.
2. Provide users with an easy way to identify available parking spaces.
3. Develop an administrative feature for monitoring and managing parking slot information.
4. Maintain organized records of parking activities and slot usage.
5. Reduce the time and effort required for drivers to locate available parking spaces.
6. Create a simple and user-friendly interface that can be easily understood by both users and administrators.
7. Evaluate the proposed system in terms of functionality, usability, efficiency, and reliability.

1.4 Scope and Limitations

Scope

The study will focus on the development of a **Parking Slot and Management System** for a selected parking area. The system will primarily handle the monitoring and management of parking slot availability.

The proposed system will include features that allow users to view available and occupied parking slots. It will also provide an administrator interface for managing parking spaces and monitoring parking-related information. The system may maintain records such as parking slot status, vehicle information, and parking activity, depending on the final design of the application.

The system will be designed with a user-friendly interface to make it easier for users to understand the current parking situation. The administrator will be responsible for managing the parking information stored in the system.

Limitations

The proposed system will be limited to the selected parking area where it will be implemented and tested. It will not automatically manage parking spaces outside the designated area.

The accuracy of real-time parking availability will depend on the method used to detect or update the status of each parking slot. If sensors or other hardware components are included in the final implementation, their accuracy may be affected by hardware limitations, connectivity problems, or environmental conditions.

The system will also not be responsible for physically controlling vehicles, parking barriers, or security equipment unless these features are specifically included in the final system design.

The study will focus primarily on **parking slot monitoring and management** and will not cover advanced features such as nationwide parking integration, automatic payment processing, or integration with all existing parking facilities.

1.5 Significance of the Study

The proposed **Parking Slot and Management System** may provide benefits to different users and groups.

Drivers. The system can help drivers identify available parking spaces more easily. By providing parking slot information, it may reduce the time spent searching for an available space.

Parking Administrators. The system can provide administrators with a more organized way of monitoring parking slots. It can also help them keep track of parking information and manage the status of available spaces.

Parking Area Owners or Management. The system can help improve the organization and management of their parking facilities. The information collected by the system may also be useful in monitoring parking usage and identifying possible improvements.

Researchers and Future Developers. This study may serve as a reference for students and researchers who plan to develop similar parking management applications. Future developers may also use the proposed system as a starting point for adding features such as online reservation, automated payment, sensor-based monitoring, or mobile notifications.

Users and the Community. A more organized parking system can contribute to a smoother parking experience by reducing unnecessary searching and movement of vehicles within the parking area.

1.6 Definition of Terms

For better understanding of the study, the following terms are defined according to their use in the proposed system:

Administrator – The person responsible for managing the parking system, including parking slot information and monitoring parking activities.

Available Slot – A parking space that is currently vacant and can be used by a vehicle.

Database – An organized collection of information used by the system to store parking-related data and records.

Occupied Slot – A parking space that is currently being used by a vehicle.

Parking Management – The process of monitoring, organizing, and managing the use of parking spaces.

Parking Slot – A designated space where a vehicle can be parked.

Parking Slot Status – The current condition of a parking space, such as available or occupied.

Parking User – A person who uses the system to view or access information about parking spaces.

Real-Time Monitoring – The process of providing updated information about the current status of parking slots.

System – Refers to the proposed Parking Slot and Management System developed in this study.

User Interface (UI) – The part of the application through which users interact with and access the parking system.